

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Mining Engineering

2. Subject Name: Mineral Economics and Business **L-T-P:** 3-0-0 **Credits:** 3

3. Pre-requisites: Development of Mineral Deposits

4. Syllabus and reference books:

Syllabus:

- Mineral industry in India: Public and Private Entrepreneurship;
- Organisation of mining enterprises;
- Mineral resources and reserves, Mineral resource estimation technics;
- United Nations Framework Classification for Resources (UNFC) and JORC classification, Techno-economic feasibility studies;
- Cut-off grade, dilution and recovery, mineral conservation and development;
- Time value of money, Discounted cash-flow concept, Net present value and Internal Rate of Return (IRR), Determining appropriate discount rate;
- Mine valuation, Project financial indicators;
- Consumer price index, Effects and inflation, Real and nominal interest rates;
- Cost-accounting in mining operations, Cost-volume-profit analysis, Depreciation and Amortization;
- Introduction to mining finance, non-banking mine finances and stock operations; Quantification and management of mining investment risk.
- Coal grade and pricing, Average Sale Price (ASP) fixing for metalliferous ores
- Mining lease, Mineral auction, Cost of mine closure;
- Taxation system applied to the mining industry, National Mineral Policy;
- Introduction to mineral trade and business management.

Reference Books:

1. Gentry D.W., O'Neil T.J. (1984). Mine Investment Analysis. Society for Mining Metallurgy & Exploration, Englewood, CO (USA).
2. Ray S. C., Sinha I. N. (2016). Mine and Mineral Economics, PHI Learning.
3. Runge I. C. (1998). Mining Economics and strategy. Society for Mining Metallurgy & Exploration, Englewood, CO (USA).
4. Michael S. (2019). Mining Capital, Springer Nature, Switzerland AG.
5. Marian R., Wårell L. (2021). A Handbook of Primary Commodities In The Global Economy, Cambridge University Press.
6. Prasanna C. (2019). Financial Management: Theory and Practice (10th Edition), McGraw Hill Education (India).

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Mineral industry in India: Public and Private Entrepreneurship; Organisation of mining enterprises;	3
2.	Mineral resources and reserves, Mineral resource estimation technics; United Nations Framework Classification for Resources (UNFC) and JORC classification, Techno-economic feasibility studies;	4
3.	Cut-off grade, dilution and recovery, mineral conservation and development;	4
4.	Time value of money, Discounted cash-flow concept, Net present value and Internal Rate of Return (IRR), Determining appropriate discount rate;	4
5.	Mine valuation, Project financial indicators;	2
6.	Consumer price index, Effects and inflation, Real and nominal interest rates;	3
7.	Cost-accounting in mining operations, Cost-volume-profit analysis, Depreciation and Amortization;	4
8.	Introduction to mining finance, non-banking mine finances and stock operations; Quantification and management of mining investment risk.	4
9.	Coal grade and pricing, Average Sale Price (ASP) fixing for metalliferous ores	2
10.	Mining lease, Mineral auction, Cost of mine closure;	2
11.	Taxation system applied to the mining industry, National Mineral Policy;	2
12.	Mineral trade and business: national and international perspectives;	2
Total number of hours		36